

BARATH BALAMURUGAN

Boston, MA | +1 857-398-8279 | barath.balamurugan@outlook.com | [LinkedIn](#) | [GitHub](#) | [barathb.com](#)

EDUCATION

Northeastern University, Boston

Sep 2024 – Dec 2026

Masters in Robotics with concentration in Computer Science | GPA – 4.00/4.00

Relevant Coursework: Robot Mechanics & Control, Supervised Machine Learning, Pattern Recognition & Computer Vision, Reinforcement Learning

Sri Krishna College of Engineering and Technology, India

Jun 2019 – Mar 2023

Bachelor of Engineering in Mechatronics - CGPA – 9.07/10

Relevant Coursework: Mobile Robotics, Autonomous Vehicle Guidance System, Virtual Reality, Embedded System for Mechatronics, Internet of Things for Mechatronics, Principles of AI and Expert Systems.

PROFESSIONAL EXPERIENCE

Research and Development Coop | Berkshire Grey

Jan 2026 – Present

- Developed a human-in-the-loop teleoperation feature to recover unsupported or failed robotic picks, increasing manipulation robustness for edge-case objects.
- Developed scripts and a test methodology to evaluate robotic-picking setup performance, identify failure modes, refine system configurations, improve grasp accuracy, and validate trigger conditions.
- Calibrated OptiTrack systems and investigated VRPN/ROS 2 clock domains, validating timestamp sources for accurate sensor-data synchronization across system components.

Research Assistant | Northeastern University

Sept 2024 – Present

- Built and demonstrated an end-to-end XR communication pipeline, integrating hardware sensors and real-time visualization; implemented a multi-sensory VR training module validated through pilot user sessions.
- Developing a physical probe that streams sensor data to VR application for live intraoperative updates; prototypes the device and validated real-time ingestion in VR with stable update rates under lab conditions.
- Designing XR/VR training workflows augmented with haptics and other sensory feedback to improve skill transfer and reduce training time.

Project Engineer | Soliton Technologies

Jun 2023 – July 2024

- Architected and optimized LabVIEW and Python APIs, reducing validation time by 35% across client hardware testing projects.
- Formulated and deployed 5+ custom utilities, enhancing testing accuracy and cutting manual errors by 40%.
- Conducted over 10+ hardware integration tests to validate API performance in diverse client environments.

PROJECTS

XR Enabled Robot Teleoperation and Digital Twin System | Unity, ROS, Python, C#, Blender, RPi, Jetson

Sept 2024 – Aug 2025

- Developed a VR application in Unity using C# to simulate and teleoperate a 6 DoF robotic manipulator running on Jetson.
- Integrated ROS for real-time communication, utilizing ROS2 topics and ROS TCP Connector for seamless data exchange.
- Implemented Edge AI (RPi) to process camera streams at the edge via ROS2. The model performs real-time feature detection for environmental awareness and feeds structured outputs back to the user for further processing and decision support.

Autonomous Dice Sorting | ROS, YOLO, MoveIt, Jetson

March 2025 – April 2025

- Designed an end-to-end autonomous sorting system combining real-time YOLO detection with depth sensing to localize dice and command pick and place via MoveIt.
- Created a dataset (Blender renders, real images, augmented data) and trained for 100 epochs, reached 91.3% accuracy.
- Reduced integration risk by modularizing perception (YOLO, depth) and control (MoveIt, ROS), enabling portable deployment on low-power hardware.

AR-Surgery | Digital Twin for Surgical Guidance | Apple Vision Pro, ESP, SwiftUI, RealityKit, ARKit

April 2025 - Present

- Prototyped a sensorized probe streaming live telemetry into an AR Surgical guidance app that aligns 3D patient/phantom models with real anatomy for immersive intraoperative visualization.
- Implemented spatial registration and object tracking pipelines improving overlay precision and procedural reliability.
- Designed surgeon-centric AR interfaces for line feedback and training in complex surgical workflows.

PUBLICATIONS

- Remote Human-Robot Collaboration in XR. In Proceedings of the 26th International Workshop on Mobile Computing Systems and Applications (HotMobile '25). [<https://doi.org/10.1145/3708468.3715687>] | Best Demo Award.
- XRfab: Immersive Cleanroom Training with Digital Twins and XR for Semiconductor Manufacturing | IEEE International Symposium on Emerging Metaverse (ISEMV '25). | [Accepted for Publication]
- Actuation of Braille Text into Braille Code and Braille Board with Navigation System | IEEE Xplore (Institute of Electrical and Electronics Engineers Xplore) | 2023 – [<https://ieeexplore.ieee.org/document/10568883>].

PROFESSIONAL SKILLS

Languages : Python, C++, C#, Swift, SwiftUI, Java, R, JavaScript, HTML, CSS, LabVIEW

Frameworks & Libraries : ROS2, React, React Native, PyTorch, Scikit-learn, OpenCV

Software & Tools: Unity, Xcode, Fusion360, AutoCAD, SolidWorks, Reality Composer Pro, Android Studio, Blender, MATLAB, JIRA, GitHub, SVN

Platforms : Meta Quest, Apple Vision Pro, VR Development, Android Development

Hardware : Jetson, Raspberry Pi, ESP